

Naturally developing memory T cell xenoreactivity to swine antigens in human peripheral blood lymphocytes.



Naturally developing xenospecific Abs are well-documented barriers to xenograft transplantation in humans, but whether analogous xenoreactive T cell immunity develops is not known. We used an enzyme-linked immunospot assay to determine the frequency and cytokine profiles of xenoreactive PBLs from a panel of human volunteers. Because naive T cells produce only IL-2 in short term culture, IFN-gamma production by this approach is a measure of a memory immune response. Stimulation of human PBLs or purified T lymphocytes with stimulator cells from inbred swine revealed a high frequency of IFN-gamma producers with 5-fold fewer IL-2 producers. In contrast, lymphocytes obtained from neonatal umbilical cord blood contained swine-specific IL-2 producers but few IFN-gamma producers, which is what one would expect to find with a naive phenotype. Moreover, PBLs from adults with a history of abstention from pork consumption responded to swine cells with a significantly lower frequency of IFN-gamma producers than PBLs from adults with unrestricted diets did, suggesting that pork consumption may result in priming of swine-specific T cell immunity. Our findings provide the first evidence for naturally occurring xenospecific T cell immunity in humans. The detected strength of this memory response suggests that it will present a formidable barrier to transplantation of swine organs.